

L21: Generating Functions

CS1010 Discrete Mathematics

31st Oct 2023 (Tue)

Recap and Today's plan

- ~~Last time~~
 - ~~Inclusion-Exclusion~~
- ~~Today:~~ *Last time*
 - Recurrences, Generating Functions
 - Generating functions for counting

Solving Recurrences via generating functions

Sequences

- A recurrence relation defines a relation between terms of an (infinite) sequence
 - E.g. $f_n = f_{n-1} + f_{n-2}$
- Express the n -th term in terms of previous terms of the sequence
- One technique to solve recurrences is to write a *generating function* for the sequence

From Sequences to Generating Functions

- Given an infinite sequence $a_0, a_1, \dots$, define the generating function

$$A(x) := \sum_{i=0}^{\infty} a_i x^i = a_0 + a_1 x + a_2 x^2 + \dots$$

- Example: if $\forall i \ a_i = 1$, then $A(x) = 1 + x + x^2 + \dots = \frac{1}{1-x}$
 - $A(x) = 1 + x + x^2 + \dots = \frac{1}{1-x}$
 - The expression on the right is called a *closed form* expression for the GF

- What about the sequence 1,0,1,0, ...?

$$A(x) = 1 + x^2 + x^4 + \dots = \frac{1}{1-x^2}$$

Finding solutions

- Coefficient of x^n in $A(x)$ is a_n

$$b_n = \underline{b_{n-1}} + b_{n-2} .$$

$$\phi(x) = \sum_{i \geq 0} b_i x^i$$

↓

Find a closed-form for this $\phi(x)$

↓

Coeff of x^n in $\phi(x)$ is b_n .

- Finding the coefficient of x^n will require an extension of the usual binomial theorem.

Extended Binomial coefficients

- For any $u \in \mathbb{R}$: $\binom{u}{k} := \begin{cases} \frac{\overbrace{u(u-1)\dots(u-k+1)}^{\text{product of } k \text{ terms}}}{k!} & \text{if } k > 0 \\ 1 & \text{if } k = 0 \end{cases}$

- Example: $\binom{-2}{3} = \frac{(-2)(-3)(-4)}{3!} = -4$

- $\binom{\frac{1}{2}}{3} = \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)}{3!} = \frac{1}{16}$

Extended Binomial Theorem

- Let $x \in \mathbb{R}$ with $|x| < 1$. Then

$$(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k$$

$u \in \mathbb{R}$ ∞ extended bin coeff seen last slide.

- E.g. for $u = -1$

$$(1+x)^{-1} = \sum_{k=0}^{\infty} \binom{-1}{k} x^k$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 \dots$$

- A very useful fact: $\binom{-k}{r} = (-1)^r \binom{k+r-1}{r}$ $\binom{-1}{k} = \frac{(-1)(-2)\dots(-k)}{k!} = (-1)^k$

Proof: $\frac{(-k)(-k-1)(-k-2)\dots(-k-(r-1))}{r!}$

$$= (-1)^r \frac{k(k+1)\dots(k+r-1)}{r!} = (-1)^r \binom{k+r-1}{r}$$

Solving Recurrences

A simple recurrence

- $t_n = 2t_{n-1}$
 - $t_0 = 1$

- Closed form: $t_n = 2^n$

- Using Generating functions:

$$t_0, t_1, t_2, \dots$$

$$T(x) := t_0 + t_1 x + t_2 x^2 + \dots$$

$$\frac{2xT(x) = 2t_0 x + 2t_1 x^2 + \dots}{T(x) - 2xT(x) = t_0 + (t_1 - 2t_0)x + (t_2 - 2t_1)x^2 + \dots}$$

$$T(x) - 2xT(x) = t_0 + (t_1 - 2t_0)x + (t_2 - 2t_1)x^2 + \dots$$

$$\Rightarrow T(x) - 2xT(x) = t_0 = 1.$$

$$\Rightarrow T(x)(1 - 2x) = 1.$$

$$\Rightarrow T(x) = \frac{1}{(1 - 2x)}$$

t_n is the coeff of x^n in $T(x)$. $\rightarrow \underbrace{[x^n] T(x)}_{\text{notation}}$

$$T(x) = \frac{1}{(1-2x)}$$

$$t_n = [x^n] (T(x)) = [x^n] \frac{1}{(1-2x)}$$

$$\frac{1}{1-2x} = 1 + 2x + (2x)^2 + \dots$$

$$\therefore \text{Coeff of } x^n \text{ in above} = 2^n.$$

$$\therefore t_n = 2^n.$$

General Recipe

- Given a recurrence $t_n = 2t_{n-1}$
- Write a generating function for the sequence $T(x) = t_0 + t_1x + \dots$
- Find a closed form of the generating function *without involving t_i 's*
 ↳ Use recurrence relation
- Find the coefficient of x^n in the closed form expression (using binomial theorem, or otherwise), this will be t_n

Solving the Fibonacci Recurrence

- $f_n = f_{n-1} + f_{n-2} \quad \forall n \geq 2,$
- $f_0 = 1, f_1 = 1$

$$\phi(x) (1 - x - x^2) = f_0 = 1$$

$$\Rightarrow \phi(x) = \frac{1}{(1 - x - x^2)}$$

- Let $\phi(x) = f_0 + f_1 x + f_2 x^2 + \dots$

- Consider $x\phi(x) = \sum_i f_i x^{i+1}, x^2\phi(x)$

$$\phi(x) = f_0 + f_1 x + f_2 x^2 + f_3 x^3 + \dots$$

$$\begin{array}{l} - \left\{ \begin{array}{l} x\phi(x) = f_0 x + f_1 x^2 + f_2 x^3 + \dots \\ x^2\phi(x) = f_0 x^2 + f_1 x^3 + \dots \end{array} \right. \end{array}$$

$$\Rightarrow \phi(x) - x\phi(x) - x^2\phi(x) = f_0 + \cancel{(f_1 - f_0)}x + (f_2 - (f_0 + f_1))x^2 + \cancel{(f_3 - (f_1 + f_2))}x^3 + \dots$$

$f_1 = f_0$

- Thus, $\phi(x) = \frac{1}{1-x-x^2} = \frac{c_1}{(1-\alpha_1 x)} + \frac{c_2}{(1-\alpha_2 x)}$

- $f_n = [x^n]\phi(x)$ (the coefficient of x^n in $\phi(x)$)

- Finding the coefficient here seems difficult, as the denominator is a polynomial in x . Can we simplify it?

$$\phi(x) = \frac{1}{(1-x(1+x))} = 1 + x(1+x) + x^2(1+x)^2 + \dots$$

Partial Fractions Lemma

Let $p(x)$ have degree $\leq n$. Then, suppose $\alpha_1, \dots, \alpha_n$ are all distinct, then there exist constants $c_1, \dots, c_n$ such that:

$$\frac{p(x)}{(1 - \alpha_1 x)(1 - \alpha_2 x) \dots (1 - \alpha_n x)} = \frac{c_1}{(1 - \alpha_1 x)} + \dots + \frac{c_n}{(1 - \alpha_n x)}$$

Back to $\phi(x)$

$$(1-x-x^2) = (1-\alpha_1 x)(1-\alpha_2 x)$$

for some α_1, α_2 .

$$\alpha_1 = -\left(\frac{1+\sqrt{5}}{2}\right), \alpha_2 = -\left(\frac{1-\sqrt{5}}{2}\right)$$

$$\bullet \frac{1}{1-x-x^2} = \frac{c_1}{(1-\alpha_1 x)} + \frac{c_2}{(1-\alpha_2 x)}$$

• Solve for c_1 and c_2 :

$$\frac{1}{(1-\alpha_1 x)(1-\alpha_2 x)} = \frac{c_1}{(1-\alpha_1 x)} + \frac{c_2}{(1-\alpha_2 x)}$$

$$1 = c_1(1-\alpha_2 x) + c_2(1-\alpha_1 x)$$

$$\text{Put } x = \frac{1}{\alpha_1}; \quad 1 = c_1 \left(1 - \frac{\alpha_2}{\alpha_1}\right)$$

$$x = \frac{1}{\alpha_2}; \quad 1 = c_2 \left(1 - \frac{\alpha_1}{\alpha_2}\right)$$

Wanted $[x^n]$ in $\frac{1}{(1-x-x^2)}$

$$= [x^n] \frac{C_1}{(1-\alpha_1 x)} + [x^n] \frac{C_2}{(1-\alpha_2 x)}$$

$$= C_1 \alpha_1^n + C_2 \alpha_2^n.$$

since

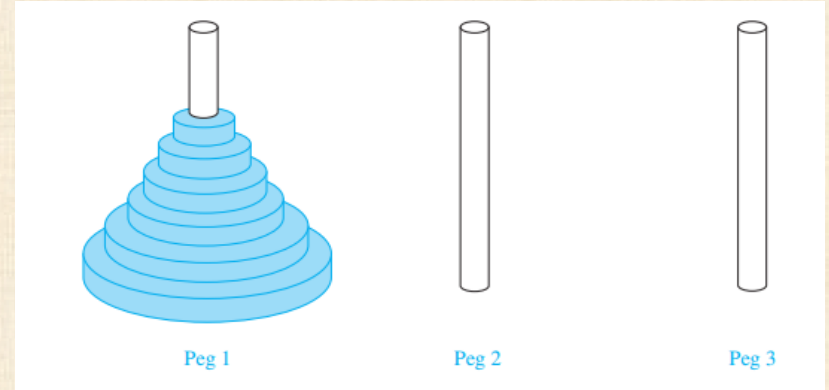
$$\left(\frac{1}{1-dx} = 1 + dx + d^2x^2 + \dots \right)$$

$$\therefore f_n = C_1 \alpha_1^n + C_2 \alpha_2^n.$$

$$\left\{ \begin{array}{l} C_1 = \frac{1}{(1-\alpha_2/\alpha_1)} \\ C_2 = \frac{1}{(1-\alpha_1/\alpha_2)} \\ \alpha_1 = -\left(\frac{1+\sqrt{5}}{2}\right) \\ \alpha_2 = -\left(\frac{1-\sqrt{5}}{2}\right) \end{array} \right.$$

The Tower of Hanoi Recurrence

- $t_n = 2t_{n-1} + 1$
 - $t_0 = 0$ $\hookrightarrow (2^n - 1)$



- How to solve this recurrence?

- Let $T(x) = \sum_i t_i x^i = t_0 + t_1 x + t_2 x^2 + \dots$

$$\frac{2xT(x)}{T(x) - 2xT(x)} = \frac{2t_0 x + 2t_1 x^2 + \dots}{t_0 + (t_1 - 2t_0)x + (t_2 - 2t_1)x^2 + \dots}$$

$$\begin{aligned} T(x) - 2xT(x) &= t_0 + (t_1 - 2t_0)x + (t_2 - 2t_1)x^2 + \dots \\ &= 0 + \underbrace{x + x^2 + \dots}_{= \frac{x}{1-x}} \end{aligned}$$

$$T(x) (1-2x) = \frac{x}{(1-x)}$$

$$T(x) = \frac{x}{(1-x)(1-2x)}$$

$$t_n = [x^n] T(x) = [x^{n-1}] \frac{1}{(1-x)(1-2x)}$$

$$\frac{1}{(1-x)(1-2x)} = \frac{2}{(1-2x)} - \frac{1}{(1-x)}$$

$$\therefore t_n = \text{coeff of } x^{n-1} \text{ in } \underbrace{\frac{2}{(1-2x)}}_{2 \times 2^{n-1}} - \frac{1}{(1-x)}$$

$$\therefore t_n = 2^n - 1.$$

$$T(x) = t_0 + t_1 x + t_2 x^2 + \dots$$

$$t_n = [x^n] T(x)$$

Solving a more general linear recurrence

- $a_n = 2a_{n-1} + a_{n-3}$

$$\begin{array}{rcl} A(x) & = & a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots \\ - (2x A(x) & = & 2a_0x + 2a_1x^2 + 2a_2x^3 + \dots) \\ - (x^3 A(x) & = & a_0x^3 + a_1x^4 + \dots) \\ \hline \end{array}$$

$$A(x) - 2x A(x) - x^3 A(x) = a_0 + (a_1 - 2a_0)x + (a_2 - 2a_1)x^2$$

$$A(x)(1 - 2x - x^3) =$$

↓
Find closed form, use partial fraction.

Balanced Parenthesis: Catalan recurrence

- Consider a product of $n + 1$ terms $x_0 x_1 \dots x_n$
- Add n pairs of parenthesis. E.g. For 3 numbers ($n = 2$):
 $(x_0 x_1)(x_3)$, or $(x_0)(x_1 x_3)$
For $n=3$:

$$\begin{array}{lll} ((x_0 \cdot x_1) \cdot x_2) \cdot x_3 & (x_0 \cdot (x_1 \cdot x_2)) \cdot x_3 & (x_0 \cdot x_1) \cdot (x_2 \cdot x_3) \\ x_0 \cdot ((x_1 \cdot x_2) \cdot x_3) & x_0 \cdot (x_1 \cdot (x_2 \cdot x_3)). & \end{array}$$

- This is the same as computing the product of the $n + 1$ numbers in a certain way, given a box that can only multiply 2 numbers at a time.
- In how many ways can we compute the product?

Catalan recurrence

- Let C_n be the number of ways to parenthesize $x_0 x_1 \dots x_n$.

Note that there are $n + 1$ numbers here.

- $C_n = \sum_{i=0}^{n-1} C_i C_{n-i-1}$

$$C_n = C_{n-2} + 4C_{n-1} + \dots$$

- The recurrence is *non-linear*, so finding the closed form is slightly tricky!

$$C(x) = C_0 + C_1 x + C_2 x^2 + \dots$$

Solving the Catalan Recurrence

- $C_n = \sum_{i=0}^{n-1} C_i C_{n-i-1}$

- Let $C(x) = \sum_i C_i x^i$

$$C(x) \times C(x) = \underbrace{(C_0 + C_1 x + C_2 x^2 + \dots)}_{\downarrow} (C_0 + C_1 x + C_2 x^2 + \dots)$$

$$\text{coeff of } x^k = \sum_{\substack{i,j: \\ i+j=k}} C_i C_j$$

$$x C(x)^2 = \sum_{n=1}^{\infty} \left(\sum_{i=0}^{n-1} C_i C_{n-i-1} \right) x^n = \sum_{n=1}^{\infty} C_n x^n.$$

$$\therefore x C(x)^2 = C(x) - 1.$$

$$xC^2 - C + 1 = 0$$

$$C(x) = \frac{1 \pm \sqrt{1-4x}}{2x}.$$

$$\downarrow$$
$$C(x)$$

Let $i' = i+1$:

$$x C(x)^2 = \sum_{i'=1}^{\infty} \left(\underbrace{\sum_{j=0}^{i'-1} (C_{i'-j-1} C_j)}_{C_{i'}, \text{ by recurrence relation.}} \right) x^{i'}$$

$$= C(x) - C_0 = 1.$$

$$\Rightarrow x C(x)^2 = C(x) - 1.$$

$$x C^2 - C + 1 = 0 \quad \nearrow \text{writing } C(x) \text{ as } C$$

$$\Rightarrow C(x) = \frac{1 \pm \sqrt{1 - 4x}}{2x}.$$

So, we have two possibilities:

$$C(x) = \frac{1 + \sqrt{1-4x}}{2x} \quad \text{or} \quad C(x) = \frac{1 - \sqrt{1-4x}}{2x}.$$

We should have $\lim_{x \rightarrow 0} C(x) = 1$, this is only true for $C(x) = \frac{1 - \sqrt{1-4x}}{2x}$

$$\therefore C_n = \text{coeff of } x^n \text{ in } \frac{1 - \sqrt{1-4x}}{2x}$$

$$= \text{coeff. of } x^{n+1} \text{ in } \left(-\frac{1}{2}\right) \sqrt{1-4x}.$$

Beyond this point is just simplification of the extended binomial coefficient.

- We thus want $[x^{n+1}]$ in $(1 - 4x)^{\frac{1}{2}}$
- By extended binomial theorem, this is exactly

$$\binom{\frac{1}{2}}{n+1} (-4)^{n+1}$$

- Simplify the above:

$$\begin{aligned} \binom{\frac{1}{2}}{n+1} &= \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)\dots\left(\frac{1}{2}-n\right)}{(n+1)!} = \frac{1}{2^{n+1}} \cdot \frac{1(1-2)(1-4)\dots(1-2n)}{(n+1)!} \\ &= \frac{1}{2^{n+1}} \cdot \frac{(1 \cdot 3 \cdot 5 \dots (2n-1))}{(n+1)!} (-1)^n \\ &= \frac{1}{2^{n+1}} \cdot \frac{(1 \cdot 3 \cdot 5 \dots (2n-1))}{(n+1)!} \cdot \frac{2 \cdot 4 \cdot 6 \dots 2n}{2^n n!} \cdot (-1)^n = \frac{1}{2 \cdot (n+1) 4^n} \cdot \frac{(2n)!}{n! n!} \\ &= \frac{(-1)^n}{2 \cdot (n+1) 4^n} \binom{2n}{n} \end{aligned}$$

- Thus, $[x^{n+1}]$ in $(1 - 4x)^{\frac{1}{2}}$ is:
- $$\binom{\frac{1}{2}}{n+1} (-4)^{n+1} = \frac{(-1)^n}{2 \cdot (n+1) 4^n} \binom{2n}{n} (-4)^{n+1} = -\frac{2}{n+1} \binom{2n}{n}$$
- Finally, the coefficient $C_n = \frac{(-1)}{2} \times \text{above term}$ (since we had $-\frac{\sqrt{1-4x}}{2x}$), which gives us that $C_n = \frac{1}{n+1} \binom{2n}{n}$
- (check: [Catalan number – Wikipedia](#) for more properties on the Catalan numbers)

More counting using
generating functions

Back to choosing sweets

- k types of sweets. Need to choose n total sweets. What is the number of ways of doing this?
 - Number of solutions to $x_1 + \cdots + x_k = n$ --- $(\star)$
- Consider the polynomial $1 + x + x^2 + \cdots$
 - The powers of x correspond to choices available for x_1
 - The number of solutions to $(\star)$ is the same as the coefficient of x^n in the polynomial

• $p(x) = \underbrace{(1 + x + x^2 + \dots)}_{\text{GF for } x_1} \underbrace{(1 + x + x^2 + \dots)}_{\text{GF for } x_2} \dots \underbrace{(1 + x + x^2 + \dots)}_{\text{GF for } x_k}$

$\# \text{ solutions} = [x^n] p(x).$

$p(x) = \frac{1}{(1-x)^k}$

Computing the coefficient

- For the given polynomial, $p(x) = \frac{1}{(1-x)^k} = \underbrace{(1-x)^{-k}}_{\sum_{r \geq 0} \binom{-k}{r} x^r \cdot (-1)^r}$.
- The coefficient of x^n in this can be found using the extended binomial theorem

$$\begin{aligned} \text{coeff of } x^n &= \binom{-k}{n} (-1)^n. \\ &= (-1)^n \binom{n+k-1}{n} (-1)^n = \binom{n+k-1}{n} \end{aligned}$$

Number of solutions of $\sum_{i \in [k]} x_i = n$

- We wanted to know the coefficient of x^n in $(1 - x)^{-k}$
- By extended binomial theorem, the coefficient of x^n is:
- $(-1)^n \binom{-k}{n} = (-1)^n \times (-1)^n \binom{n+k-1}{n} = \binom{n+k-1}{k-1}$

Choosing Fruits

- How many ways are there to choose n fruits so that:

- The number of Apples are even
- Bananas are a multiple of 5
- At most 4 oranges are chosen
- At most 1 Pear is chosen

$$x_1 + x_2 + x_3 + x_4 = n$$

s.t. x_1 is even.

$$5 \mid x_2$$

$$x_3 \leq 4, \quad x_4 \leq 1$$

for apples. $A(x) = x^0 + x^2 + x^4 + x^6 + \dots = \frac{1}{(1-x^2)}$

$$B(x) = x^0 + x^5 + x^{10} + \dots = \frac{1}{(1-x^5)}$$

$$O(x) = x^0 + x^1 + x^2 + x^3 + x^4 = \frac{1-x^5}{(1-x)}$$

$$P(x) = x^0 + x^1 = (1+x)$$

want $[x^n] (A(x) B(x) O(x) P(x))$

want $[x^n] p(x)$ where $p(x) = \frac{1}{(1-x^2)} \times \frac{1}{(1-x^5)} \times \frac{(1-x^5)}{(1-x)} \times (1+x)$

$$= \frac{1}{(1-x)^2}.$$

$$\therefore [x^n] p(x) = (-1)^n \binom{-2}{n} = (n+1)$$

✓

$$[x^n] \left(\frac{1}{(1-x)^{+2}} \right) = (-1)^n \binom{-2}{n} = n+1$$

Sum of digits=25

- What is the number of positive integers $x \leq 9999$ such that the sum of the digits is 25?

Number is of the form $x_1 x_2 x_3 x_4$, $0 \leq x_i \leq 9$

$\therefore$ Answer is #solns to $x_1 + x_2 + x_3 + x_4 = 25$

GF for each x_i : $1 + x + \dots$

$$0 \leq x_i \leq 9$$

$$x^9 = \frac{1-x^{10}}{1-x}$$

Want $[x^{25}]$ in $\left(\frac{1-x^{10}}{1-x}\right)^4 \equiv (1-x^{10})^4 (1-x)^{-4}$

$x \leq 99999$

$$(1-x^{10})^{+4} = \sum_{0 \leq i \leq 4} \binom{4}{i} (-1)^i x^{10i} \quad (\text{Bin thm})$$

$$(1-x)^{-4} = \sum_{j \geq 0} \binom{-4}{j} (-1)^j x^j \quad (\text{Ext Bin thm})$$

$$\text{Want } [x^{25}] \text{ in } = \sum_{0 \leq i \leq 4} \sum_{j \geq 0} \binom{4}{i} \binom{-4}{j} (-1)^{i+j} x^{10i+j}$$

$$\begin{array}{lll} x^{25} : i=0 & , j=25 & \rightarrow \binom{4}{0} \binom{-4}{25} (-1)^{25} + \\ i=1 & , j=15 & \rightarrow \binom{4}{1} \binom{-4}{15} (-1)^{16} + \\ i=2 & , j=5 & \rightarrow \binom{4}{2} \binom{-4}{5} (-1)^7 + \end{array}$$